THE UNIVERSITY OF WESTERN AUSTRALIA

Introduction to Agile Web Development

CITS3403 and CITS5505 - Agile Web Development

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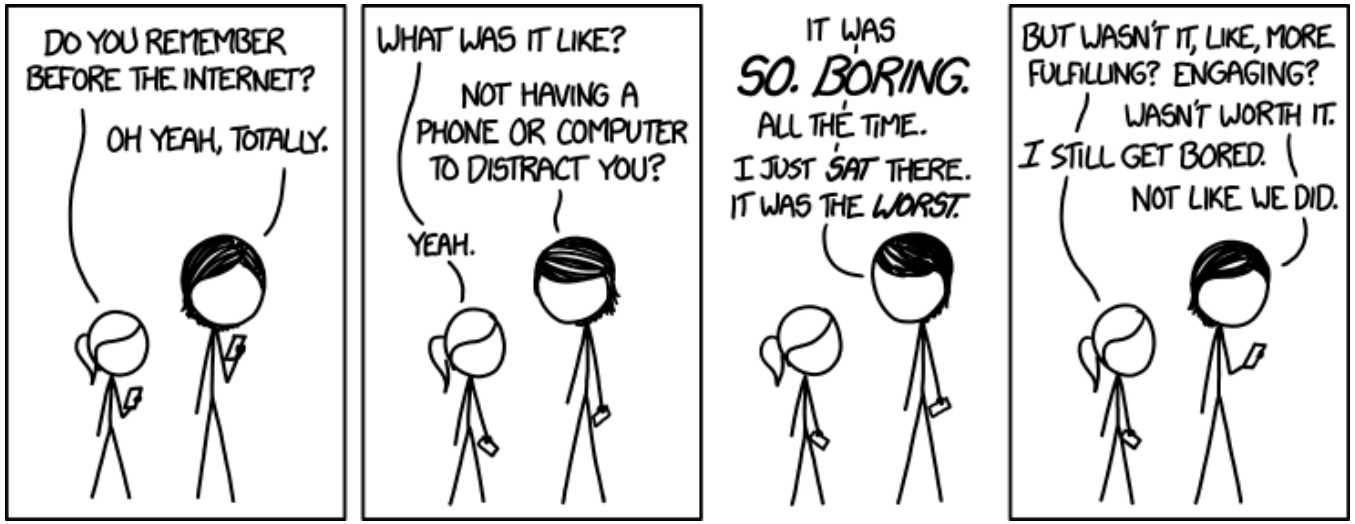
2026 - Semester 1

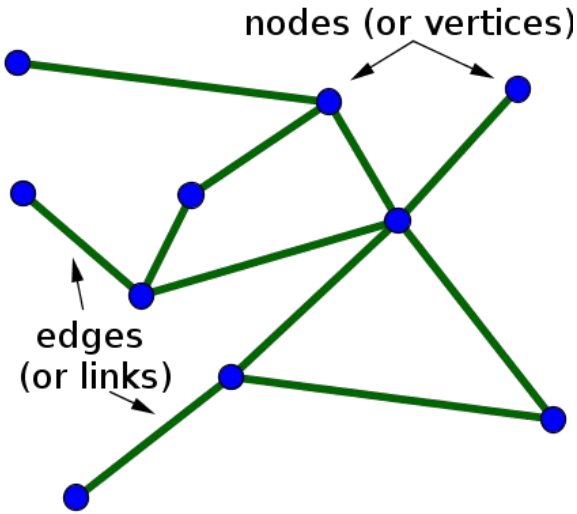
A (very) brief history of the Internet



https://xkcd.com/1348/

What is a computer network?

- A **network** is a structure linking devices together for the purpose of communication and can be modelled as a graph.



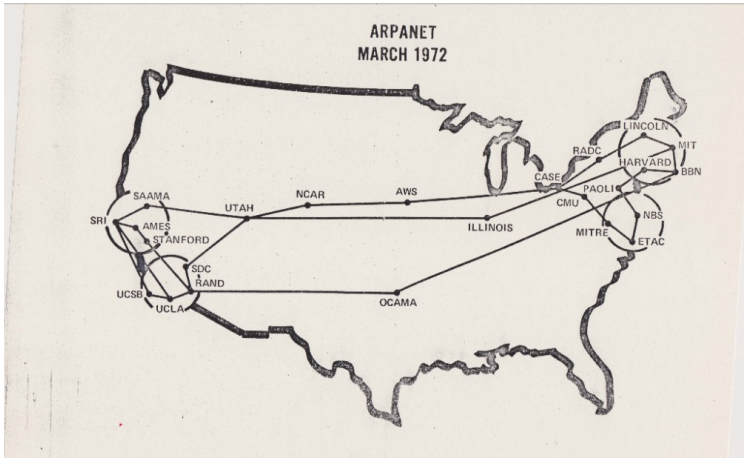
- Each individual device is modelled as a node in the graph.
- The physical connection between two devices is modelled as an edge in the graph.

Networks before the Internet

- In the US, the DoD created the Advanced Research Projects Agency (ARPA, now DARPA) in 1958. ARPA was interested in creating networks that:
  - allowed communications, program sharing, remote computer access.
  - were robust and would continue to work if some nodes “taken out” by malicious forces.

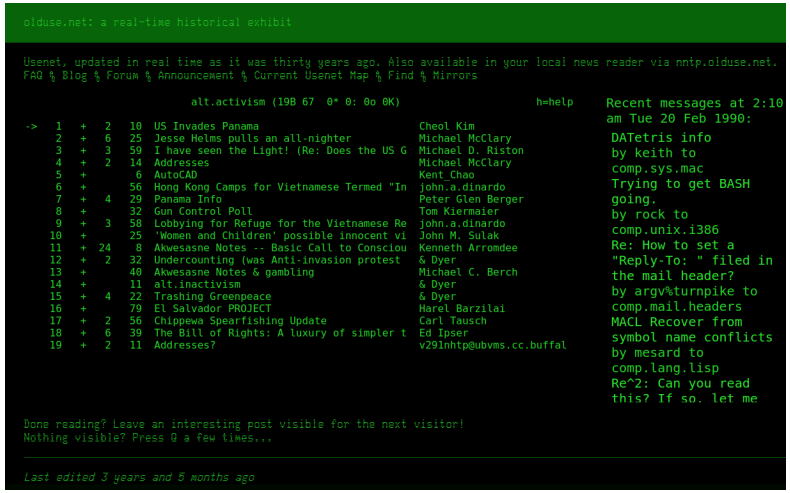
- This resulted in the ARPAnet in the late 1960s and early 1970s
  - linked about a dozen ARPA-funded research labs and universities.
  - graduate students played a large part in its development!
  - didn’t live up to intentions - mostly text-based email and limited reach.
  - but still useful and the snowball had started to roll…

- Non-ARPA-funded Universities wanted in, so other networks were created in the late 70s and early 80s
  - BITnet (Because It’s Time Network), initially electronic mail and file transfer
  - CSnet (Computer Science Network), primarily email



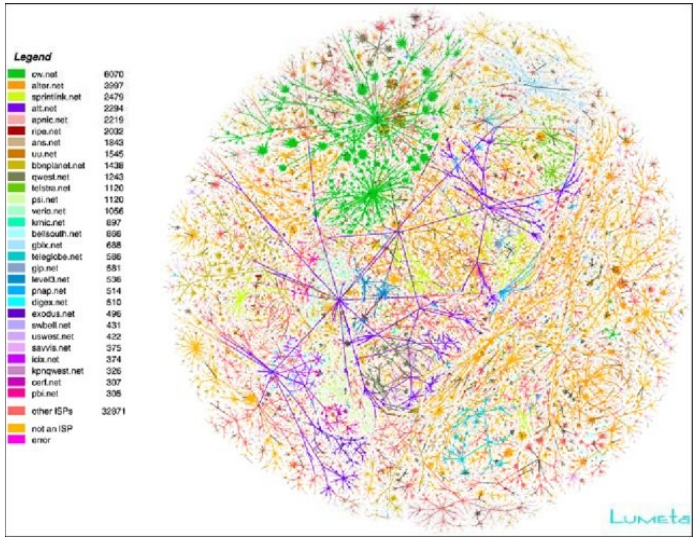
The Birth of the Internet

- NSFnet was created in 1986 by the National Science Foundation (NSF).
- Originally for non-DoD funded organisations, it initially connected five supercomputer centres but quickly spread to other academic institutions and research labs.
- By 1988/89 - began to be used commercially - mail, ISPs.
- By 1990, it had replaced ARPAnet for non-military uses and it soon became the network for everyone.
- Other networks created gateways and eventually merged with NSFNet (e.g. JANET, BITnet, Usenet, ...).
- By 1992 connected more than 1 million computers around the world and eventually became known as "the Internet".



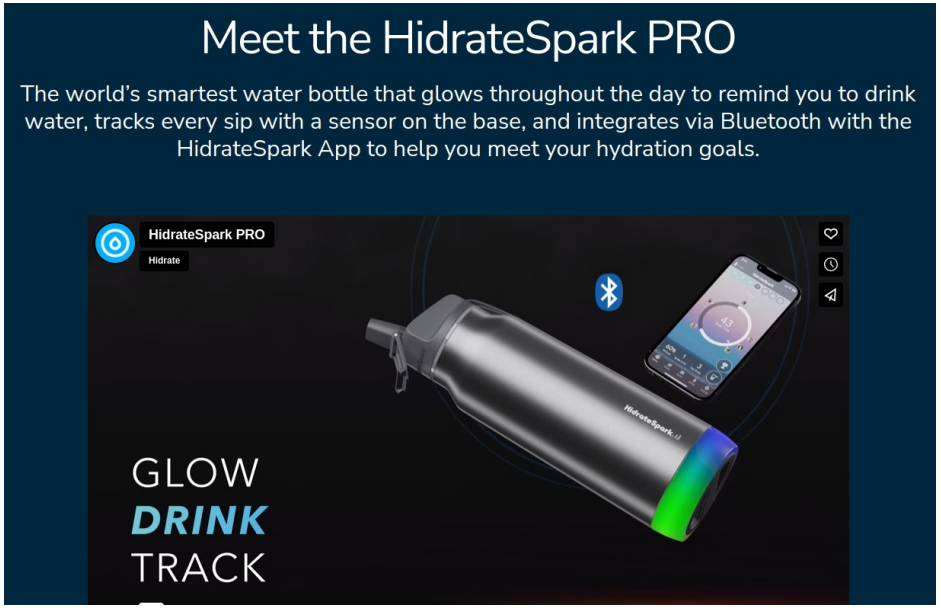
The Evolution of the Internet

- The Internet is still a network of interconnected networks.
- As it evolved high-speed backbone links were created to carry large amounts of data.
- Smaller networks connect to the backbone, enabling any user on any network to exchange data with any other user.
- Both backbone links and individual networks can be owned by companies, universities or nation states.
- It achieved ARPA's original goal that if part of its infrastructure is destroyed, data can still flow through the remaining networks (in principle).
- By the 2000s it had become indispensable global infrastructure.



The Internet Today

- Today the Internet has grown to include tens of billions of interconnected computers, smart phones, televisions, printers, fridges, watches and most crucially… *water bottles*.



- All these devices on the Internet are connected using a wide range of different types of links: copper cables, fibre-optic cables, satellites, phone lines etc.

How does the internet work?

Communication protocols

- No matter whether using fibre-optics, satellite links etc, the physical connections all send data as a sequence of bits, e.g. 101011110110100000011001.

- Network communication is possible only if computers “speak” a common language and know how to interpret the bits. These common languages are known as *protocols*.

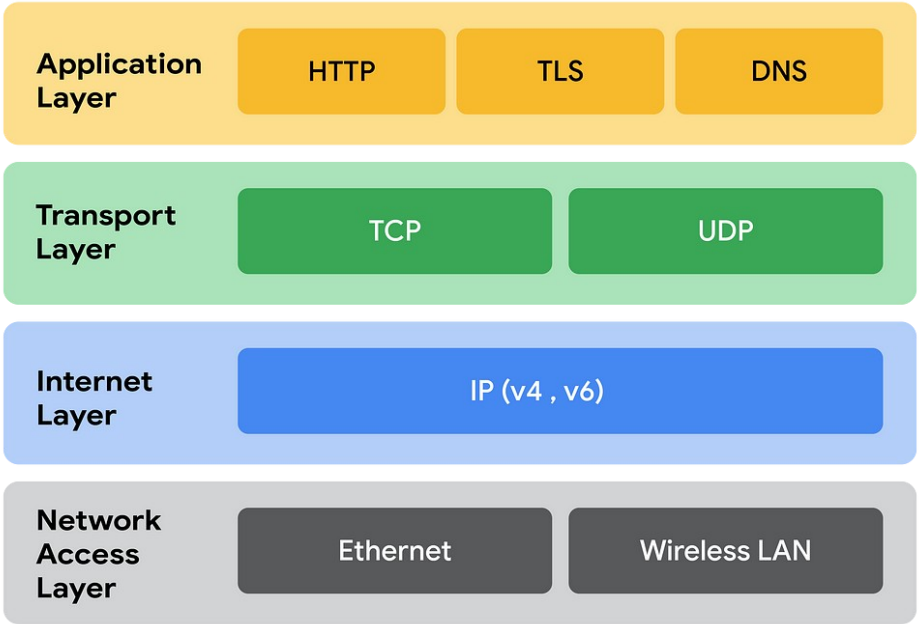
- *It's protocols all the way down:*
  - IP – consistent addressing of entities on the internet
  - BGP – finding the best routes across the internet
  - TCP – error-free delivery of streams of data
  - SMTP – sending and receiving emails.
  - FTP – sending and receiving files.
  - etc. etc.

The TCP/IP model

- Every computer and network on the Internet uses the same set of protocols - the Transmission Control Protocol/Internet Protocol, or TCP/IP for short.
- No matter what type of computer system you connect to the Internet, if it uses TCP/IP, it can exchange data with any other type of computer.
- TCP/IP was developed to guarantee the proper transmission of data, since the physical layer (e.g. wires/wi-fi/satellite) in the network may be unreliable.
- For transmission not needing guarantees, one can use User Datagram Protocol (UDP) instead of TCP. Data transmitted by UDP arrive faster, with none of the error detection or correction overheads that are in TCP/IP.
- In TCP/IP a stream of data is split into packets which are sent individually over the network.

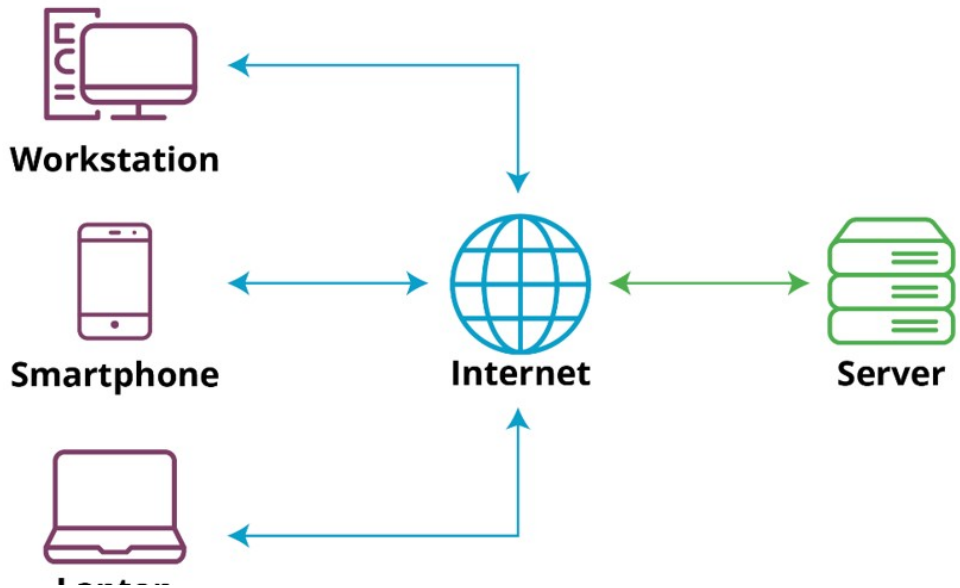
The TCP/IP model layers

- The TCP/IP model is split up into four layers:
  - The application layer protocols dictate what format the stream of data should be in for different high-level applications.
  - The transport layer protocols convert the stream of data to and from a sequence of packets and are responsible for detecting and fixing packets that are lost or corrupted during transport.
  - The internet layer protocols are responsible for transmitting a single packet from the source device to the destination device across the network.
  - The network access layer protocols describe how a single packet is transmitted across a single physical link in the network.



Client-Server Architecture

- Most communication on the internet takes the form of a client-server relationship.
  - The server is computer whose address is known, and which stores information on its file system.
  - The clients sends a request for information to the server via an agreed protocol (FTP, SMTP etc.).
  - The server transmits the requested information back to the clients.



Client-Server Architecture

- Advantages
  - Multiple clients can use a single server.
  - New clients can join the system without having to be registered in advance.
  - A single, central source of information.

- Disadvantages:
  - There is a single point of failure – the server.
  - If too many clients, the server may be overloaded with requests.

- To get around the disadvantages, we can have many duplicate servers containing the same content, at the cost that more work must be done to keep the copies of information synchronised…

Internet Applications

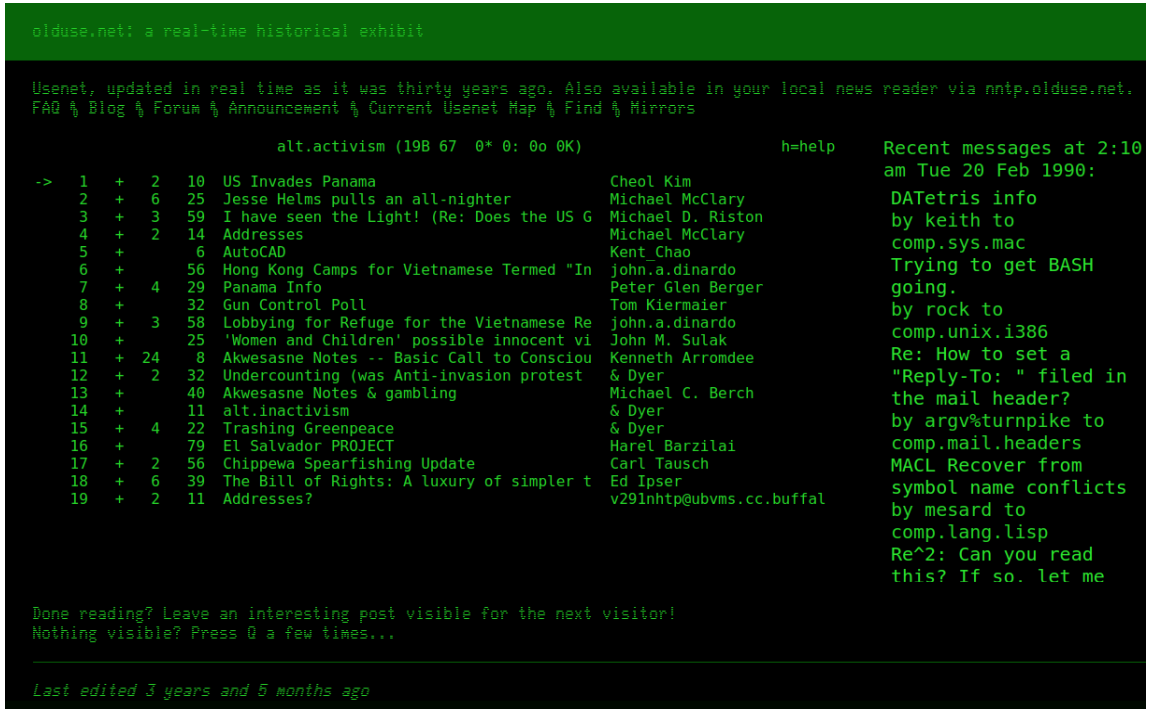
Bulletin Board System (BBS)

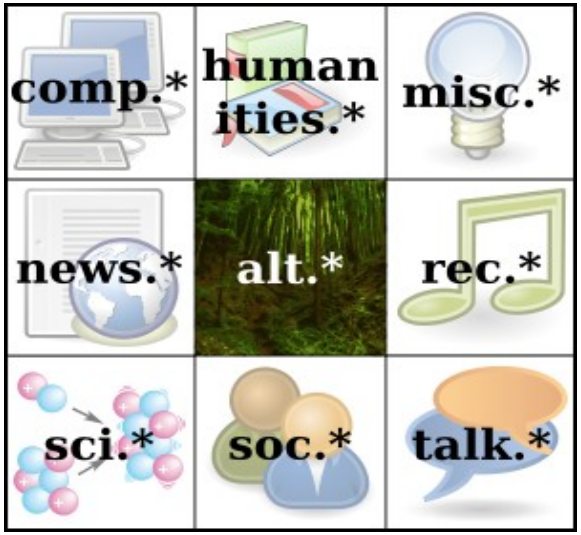
- Early interactive software, late 1970s to 1980s
- Users could login to:
  - exchange messages through mail or public message boards
  - read news and bulletins
  - upload/download software
  - play online games
- Accessed using modem and phone line
- Precursor to today's WWW

http://en.wikipedia.org/wiki/File:Monochrome-bbs.png

Usenet

- Idea conceived by Duke University grad students in 1979
- Unlike BBS, distributed network of servers (e.g. each university)
- A variety of forums called *newsgroups* (not just news - users posts)
- Threaded discussions
- Formed social communities
- Precursor to Internet forums



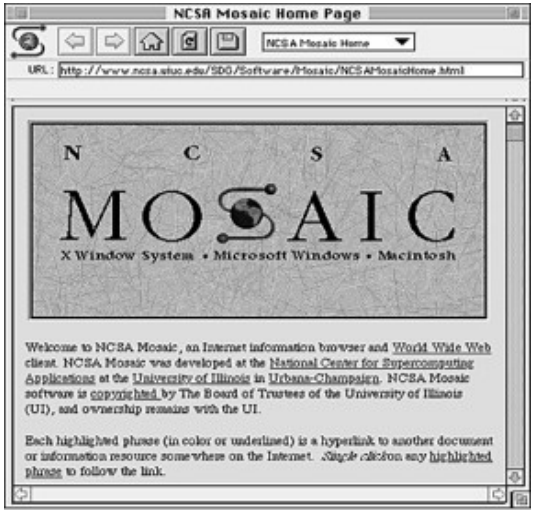


Source: Benjamin D. Esham, Wikimedia Commons
http://en.wikipedia.org/wiki/File:Usenet_Big_Nine.svg

The "killer app" - the World Wide Web

- Early internet was mainly used by people in universities and research labs.
  - Lots of information but very difficult to access.
  - You needed the IP address of the computer you wanted to connect to.
  - No uniform way of visualising the information.

- In 1991, CERN publicised new World Wide Web project, invented earlier by Tim Berners-Lee and colleagues in 1989:
  - DNS (Domain Name System) resolves domain names to IP addresses.
  - Included the first browser "WorldWideWeb" on the NeXTSTEP platform
  - First ever website still online!



- In 1993, the National Center for Supercomputing Applications (NCSA) at the University of Illinois released v1.0 of Mosaic browser:
  - written by a student, Marc Andreessen, and Eric Bina
  - first multimedia browser (mixed images and text)!

- Explosion in internet use!
  - growth of web usage in 1000s of percent
  - changed internet use forever
  - the “killer app” of the 90s

A URL (uniform resource locator)

http://www.domain.edu.au:1000/path/to/file?parameters=true#fragment

- The protocol used. Typically, *http*, *ftp*, *https*, …
- The domain name. A *domain name server* maps this to an IP address
- The port number. Servers have ports 0-65535, but HTTP defaults to port 80.
- The path (route) to the file to execute. The file is typically an HTML file, but it could also be PHP, text, PDF.
- The parameters of the request. These are specified as a set of key value pairs.
- The fragment. This anchors to a location in a page.

Client-Server Architecture of the Web

- Communication is by an agreed protocol, e.g. the HTTP (HyperText Transfer Protocol).
- The user requests a web page through the browser, a program running locally on their computer.
- The browser, as the client, locates the correct server and communicates the request.
- The server retrieves the web page from its local file system and transmits the files back to the browser.
- The browser receives the files, (usually text file containing HTML instructions) and uses them to render the web page resulting in the intricate graphics and formatting you see on your screen.

The World Wide Web today

- The world wide web is essentially the fragment of the internet accessible through web browsers.
- It is a unique engineering environment with obscure ownership and control.
- Arguably the most impressive piece of infrastructure ever built by humanity.
- Hundreds of standards have evolved that define interaction over the webpages.
- As search has come to dominate, URLs are now nearly irrelevant to users.

The World Wide Web tomorrow

New web technologies that shape future business models

Transition in web technologies

	1996 Web 1.0	2004 Web 2.0	2021 Web 3.0 (Still in development)
	The Hypertext Web	The Social Web	The Semantic Web
Features	Read-only and Static Web	Interactive	Intelligent and decentralised
Content	Static content, passive viewing	Dynamic content, platform, collaborate	AI, 3D, blockchain, web learning
Use Cases	Content and information publishing	Social Media, Microblogs, content hosting, podcasts	DeFi, tokenization of asset, online gaming, distributed ledgers, NFT, CBDC, P2P transfer
Key Technologies	Content delivery network, HTML /portals, Netscape	Mobile, Applications, Big data, AI, blockchain	Blockchain apps, advanced AI, ML, Augmented Reality, IoT, 5G
Challenges	Lacks two-way communication option for users, No algorithms	Authentication controls, data ownership, Malware, data leak, data integrity	Personal data protection, cybercrime, AI bias
Examples	CNN, BBC	Amazon, Apple, Meta, Microsoft, Twitter, Google, Tik Tok	Amazon Alexa, Wolfram Alpha, Apple's Siri , ChatGPT, Bard, ...

Source: TABInsights